

AMAZONHELP TRUST-SAFE AI SUPPORT AGENT

Track A — MVP Evaluation & Engineering Report

Submission: Hiver SDE Intern Take-Home Assignment | **Brand:** AmazonHelp | **Dataset:** Kaggle Customer Support on Twitter

Executive Summary

This report documents an offline, end-to-end AI customer-support agent built for AmazonHelp as the Track A submission for the Hiver SDE Intern take-home assignment. The system reconstructs branching Twitter support threads, classifies customer intent against a ten-category taxonomy derived directly from the data, retrieves outcome-aware historical precedents, generates grounded replies, and routes every case through a deterministic Trust Gate before automation or human hand-off.

The governing engineering principle is that a support agent's default behaviour should be to resolve what it can prove, verify what it cannot yet establish, and transfer ownership when safe resolution is unavailable — rather than to maximise fluent replies or automation volume for their own sake. Every quantitative claim in this report is drawn from a single, locked, 200-case, human-verified gold set that was excluded from training, calibration, and retrieval indexing throughout development.

The prototype demonstrates a complete, auditable decision pipeline with measurable evidence across intent classification, retrieval, safety, and capability behaviour. It also demonstrates, by design, where the evidence is not yet sufficient for a production claim — most notably reply-quality judge calibration and safety recall. Section 7 treats this distinction as a first-class deliverable rather than a footnote.

1. Problem Framing & Scope

1.1 The Assignment

The brief asks for a working, single-brand AI support agent that classifies incoming customer messages, drafts a reply grounded in how that brand has historically resolved similar issues, and decides — with a stated reason — whether a case should be auto-handled or escalated to a human. The brief is explicit that the evaluation and proof of correctness matter more than system sophistication, and that the report must state plainly what a headline number does and does not demonstrate.

1.2 What “Good” Means for AmazonHelp

- The system never claims an action was executed without verifiable evidence that it actually occurred.
- Historical replies are treated as behavioural precedent — a playbook — never as current factual truth about a live order, refund, or account.
- Frustration and repeated failure are treated as first-class signals, not noise to be smoothed over by another scripted reply.
- Ambiguous or incomplete evidence triggers one bounded Deep Analysis step rather than a guess.
- Explicit human requests and serious safety, legal, or identity-risk situations are hard overrides that no confidence score can suppress.
- Every autonomous response carries an AI disclosure, and every decision leaves an auditable trail.

1.3 Track A Scope and Deliberate Non-Goals

The implemented system is an offline MVP over historical AmazonHelp data. It covers ingestion, PII scrubbing, branch-aware thread reconstruction, intent classification, outcome-aware retrieval, grounded generation, safety and capability checks, Trust Gate routing, the 200-case gold set, the evaluation harness, and a local interactive UI. Live Twitter ingestion, backend writes, CRM mutation, persistent cross-session customer profiles, production queues and databases, and distributed infrastructure are documented as Track B concerns rather than implemented — a deliberate scope decision explained further in Section 11.

2. System Architecture

The pipeline follows a single directional flow: Customer Message → Thread Context → Intent Classification → Retrieval → Grounded Generation → Trust Gate → Auto-Handle / Deep Analysis / Human Escalation. The differentiator is not the classifier or the language model in isolation, but the decision layer that wraps a standard retrieval-augmented generation flow with evidence, capability, and safety gates.

Layer	Engineering Decision	Rationale
Thread Context	Branch-aware reconstruction with an explicit lifecycle	Prevents a linear-chain assumption from feeding incorrect context to every later stage.
Intent	Ten AmazonHelp-specific intents + UNKNOWN; multi-intent supported	Avoids forcing a label onto a case the taxonomy does not actually cover.
Retrieval	Semantic similarity $\times$ intent compatibility $\times$ observed-outcome quality	Prioritises historically useful behaviour without presenting history as current truth.
Generation	Grounded structured output with post-generation capability checks	Blocks unsupported execution claims (e.g. “your refund is processed”).
Trust Gate	Deterministic Auto / Deep Analysis / Human routing with hard overrides	Keeps the highest-risk decisions out of a purely probabilistic judgment loop.
Deep Analysis	One bounded verification or recovery attempt	Recovers missing evidence without producing an endless clarification loop.
Safety	Contextual, evidence-based risk detection	Separates genuine safety risk from ordinary frustration, profanity, or escalation.
Privacy	One-time canonical PII scrubbing before any downstream use	Prevents repeated transformation drift and downstream PII exposure.

Table 1. Design decisions by pipeline layer and the reasoning behind each.

3. Data & Gold-Set Methodology

3.1 Source Data and Preprocessing

The primary source is the Kaggle Customer Support on Twitter dataset — approximately three million tweets spanning dozens of brands. AmazonHelp was selected as the single pilot brand, and reply-chain identifiers were used to reconstruct root conversations. Preprocessing filtered AmazonHelp threads, scrubbed PII once on canonical unique tweet records (structured identifiers such as email, phone, and order or tracking numbers are masked; unstructured entities use named-entity recognition), applied thread-scoped deterministic placeholders so a repeated entity resolves consistently within a thread, and restricted scope to English-language conversations. Isolation is enforced at the root-conversation level to prevent train/evaluation leakage.

3.2 Gold-Set Construction

The evaluation set contains 200 human-verified root conversations, satisfying the assignment's 150–250 case requirement. Sampling deliberately over-weighted ambiguity, possible multi-intent structure, repeated customer frustration, escalation-prone signals, and low retrieval-coverage cases, rather than drawing a random cross-section of ordinary traffic. This choice is revisited in Section 7 because it directly affects how the headline classification number should be read. Each case is labelled for primary and secondary intent, multi-intent status, Trust Gate tier, frustration trajectory, safety risk, capability status, and a reference resolution recording the must-cover facts a correct reply would need to establish.

3.3 Evaluation Isolation

The 200 canonical gold-set root IDs are excluded from classifier training, validation, calibration, and the FAISS retrieval index. Raw and processed customer datasets were subsequently purged from Git history, reducing repository size from approximately 3.6 GiB to roughly 17 MiB; no API keys or environment files are tracked.

4. Evaluation Framework

Evaluation is organised around four pillars defined in the project's architecture, so that no single metric is asked to stand in for overall support quality. Deflection rate is explicitly excluded as a headline KPI: a case counted as “deflected” is not the same as a case that was actually resolved, and treating the two as equivalent was judged a bigger risk to customer trust than a lower automation number.

Pillar	Measurement / Status
Resolution Quality	Correctness and actionability of the reply against must-cover facts; full LLM-judge evaluation not yet complete.
Customer Experience	Frustration trajectory, repeat-loop signals, turns-to-resolution, tone, and customer effort.
Trust & Safety	Hallucination and false-promise prevention, safety overrides, capability honesty.

Pillar	Measurement / Status
Decision Quality	Three-class Auto / Deep / Human confusion analysis, with dangerous mis-routes weighted most heavily.

Table 2. The four evaluation pillars and what each is intended to catch.

Two baselines anchor the classification results: a trivial majority-class baseline, and a deliberately lightweight TF-IDF + Logistic Regression classifier. The classifier is intentionally simple because the primary engineering contribution of this project is the decision, grounding, retrieval, and safety layer that surrounds it — not the classifier itself.

5. Results

5.1 Headline Results

Measure	Result	What It Tells Us
Intent accuracy	49.5%	Overall classification correctness on the 200 gold cases.
Intent Macro F1	47.5%	Headline classification metric, averaged evenly across intents.
Trivial-baseline Macro F1	18.9%	Reference point the classifier is measured against.
Retrieval proxy P@1 / P@3	33.0% / 21.5%	Diagnostic retrieval signal; not a human-labelled relevance score.
Safety precision / recall	100% / 29.4%	No false alarms observed; recall is a material, documented gap.
Capability accuracy	86.5%	Capability-state classification accuracy on the gold set.
Dangerous Human → Auto misses	0	No case requiring human ownership was auto-routed.
AI disclosure	100%	Present on every tested autonomous response.

Table 3. Full result set on the locked 200-case gold evaluation.

5.2 Baseline Comparison

Model	Accuracy	Macro F1
Trivial baseline (majority class)	32.0%	18.9%
TF-IDF + Logistic Regression	49.5%	47.5%

Table 4. The lightweight classifier improves materially over the trivial baseline on both metrics.

5.3 Trust Gate Routing

Gold Tier	Correct / Total	Interpretation
Auto-Handle	0 / 22	Track A withholds autonomous action when live backend evidence is absent.
Deep Analysis	78 / 78	Every gold case needing bounded verification was correctly routed there.
Human Escalation	5 / 100	Correct on explicit hard-rule cases; weak separation from Deep Analysis overall.

Table 5. Three-class confusion outcome against the 200-case gold set. See Section 6.2 for interpretation of the Auto and Human columns.

6. Failure Analysis

The failure analysis is deliberately specific. The objective is not to obscure weak areas but to point at the engineering work most likely to improve real customer outcomes next.

6.1 Top Five Classifier Failure Modes

#	Observed Failure	Count	Engineering Hypothesis
1	Delivery_Delayed → UNKNOWN	40	Delivery language varies widely; classifier confidence collapses on incomplete or indirect phrasing.
2	Billing_Or_Prime_Charge → UNKNOWN	9	Long-tail wording overlaps heavily with cancellation and refund conversations.
3	Package_Missing_Or_Stolen → UNKNOWN	6	Overlaps with delayed-delivery language and incomplete thread context.
4	Device_Technical_Issue → UNKNOWN	5	Technical complaints are sparse and lexically diverse in this corpus.
5	Delivery_Delayed → Order_Cancellation	5	Customers raise cancellation as a consequence of delay, creating genuine boundary ambiguity.

Table 6. The classifier's dominant error mode is retreating to UNKNOWN rather than mis-firing into a wrong intent — a conservative failure, not a confident one.

6.2 Safety False-Negative Audit

The safety detector produced 12 false negatives. Manual review found that most were fraud, account-compromise, or stolen-package cases that a broader interpretation might label as safety concerns; under the locked design those categories are deliberately kept separate from immediate physical or legal safety risk. One case remained a genuine physical-safety miss after audit. The corrective action is not to overfit the detector to these 200 cases, but to expand contextual safety examples and re-evaluate — captured as Priority 2 in Section 10.

6.3 Trust Gate Interpretation

The confusion outcome in Table 5 reveals two facts at once. First, the gate produced zero dangerous Human → Auto mis-routes, which is the single most important safety property of a routing layer. Second, it does not yet reliably separate Deep Analysis from Human Escalation — the 0/22 Auto-Handle result reflects Track A's absence of live backend evidence rather than a ceiling on what automation could achieve with real capability access.

6.4 Retrieval and Resolution Interpretation

Proxy P@1 (33.0%) and P@3 (21.5%) are diagnostic, not definitive: the gold set does not contain human relevance labels for every retrieved precedent, so these numbers describe retrieval behaviour rather than proven relevance. Separately, historical Twitter threads cannot reliably distinguish a solved issue from a customer who simply stopped replying. The system therefore treats an Observed Outcome as a proxy signal and never equates silence with resolution — a decision that keeps the resolution metric honest at the cost of making it look more conservative than a production dashboard typically would.

7. What Is Misleading About My Headline Number

Headline: 47.5% Intent Macro F1.

This is the single most useful classifier statistic produced by the project, and it is not an end-to-end customer-support quality score. Presenting it without the context below would overstate what the evaluation actually proves.

Why It Can Mislead	What the Number Does Not Capture
Macro averaging	Long-tail class weaknesses can be hidden inside a single blended average.
Gold-set composition	The 200 cases were deliberately selected to surface difficult behaviour; they are not a production traffic distribution.
Intent ≠ resolution	A correctly classified intent can still produce a reply that is unhelpful, ungrounded, or unsafe.
Retrieval proxy	P@K here is not a human-validated relevance score.
Track A capability constraint	Zero correctly Auto-Handled cases reflects missing live backend evidence, not a production automation ceiling.
Judge gap	No human-calibrated LLM-judge score exists yet, so reply quality cannot be summarised by a validated single metric.

Table 7. Six specific ways the headline classification number could be over-read.

The more honest interpretation is that a simple classifier can materially and repeatably outperform a trivial baseline on this data, while the broader system still needs stronger evidence — a calibrated reply-quality judge, higher safety recall, a human-labelled retrieval benchmark, and live capability access — before any claim of production-grade support quality would be defensible. Making that distinction explicit is treated as a deliverable in its own right, not a caveat appended after the fact.

8. Trust & Safety by Design

- Historical support replies are treated as evidence of behaviour, never as authority about the current customer or order.
- Current verified data and capability state always outrank historical precedent when the two conflict.
- Unsupported action claims are blocked by a post-generation validation step, not merely discouraged at the prompt level.
- Explicit human requests are hard overrides; no confidence score is permitted to suppress them.
- Deep Analysis provides exactly one bounded recovery attempt before a case is transferred to a human, preventing an endless clarification loop.
- Every autonomous response carries an explicit AI disclosure.
- PII is scrubbed once, canonically, before any downstream use; customer text is never re-published unmasked.

These properties are the concrete expression of the architecture's stated design goals: grounded, capability-honest, loop-aware, trust-aware, escalation-safe, transparent, privacy-preserving, auditable, and customer-focused.

9. Known Limitations & Submission Readiness

Area	Status
200-case gold set	Complete
Intent classifier + baseline comparison	Evaluated
Retrieval diagnostics	Evaluated, with proxy-metric limitation documented
Trust Gate	Evaluated
Safety detector	Evaluated; recall limitation documented
Capability guard	Evaluated
Loop-breaker	Dedicated 5-case integration test passed
LLM-judge human calibration	Not completed
Production backend integration	Deferred to Track B by design

Table 8. Submission-readiness checklist across every evaluated component.

- Safety recall (29.4%) remains the most consequential open gap and is prioritised first in Section 10.
- Classifier uncertainty is highest in long-tail and semantically overlapping intents (Table 6).
- Retrieval evaluation is proxy-based pending a human-labelled relevance benchmark.
- LLM-judge human agreement has not yet been measured; no reply-quality score is reported as a result.
- No live backend means autonomous action is intentionally, not accidentally, constrained in Track A.
- Persistent customer memory and production escalation infrastructure are out of scope for this submission.

10. Roadmap — Priorities for the Next Iteration

With one additional week, the priority is evidence quality and the highest-risk failure modes, not additional architectural complexity.

Pri.	Next-Week Work	Expected Value
1	Complete human calibration of the LLM judge; measure agreement and inspect disagreements.	Makes reply-quality evaluation defensible for the first time.
2	Improve safety recall with audited contextual examples, without conflating frustration with safety.	Directly reduces dangerous false negatives.
3	Add hard training examples for Delivery vs. Missing, Device, Billing/Prime, and UNKNOWN boundaries.	Targets the exact failure modes observed in Table 6.
4	Hand-label a small human retrieval-relevance benchmark.	Converts the current proxy P@K into a defensible metric.
5	Wire one read-only backend capability, such as order-status lookup.	Tests the capability gate against real evidence for the first time.
6	Re-run the locked evaluation after any model or data change.	Preserves reproducibility and prevents silent metric drift.

Table 9. Roadmap ordered by expected reduction in customer-facing risk.

11. Decision Log — Fifteen Non-Obvious Decisions

#	Decision	Rationale
1	AmazonHelp as the single pilot brand.	Keeps the take-home focused and every design choice traceable to one real support context.
2	Branch-aware thread reconstruction.	The source data is a reply graph, not a clean linear conversation.
3	ACTIVE / WAITING / CLOSED lifecycle, separate from resolution outcome.	Customer silence is not treated as evidence of success.
4	Observed Outcome instead of “resolved = thread ended.”	The dataset provides no authoritative resolution ground truth.

#	Decision	Rationale
5	Historical replies treated as playbooks, never as current truth.	Prevents outdated support behaviour from being replayed as current policy.
6	Outcome-aware retrieval that still surfaces UNKNOWN/contradictory evidence.	Lets successful precedent influence ranking without hiding uncertainty.
7	UNKNOWN retained as a valid intent label.	Avoids forcing a classification the evidence does not support.
8	Multi-intent classification supported.	Customers frequently raise more than one independent issue in a single thread.
9	Intent, frustration, safety, and capability scored as separate signals.	Each answers a different operational question and should not be conflated.
10	Deep Analysis as a bounded middle tier.	Allows one meaningful recovery attempt before committing to human ownership.
11	Hard overrides for explicit human requests and serious safety/legal cases.	These must never be suppressed by a confidence score.
12	Capability honesty as an authoritative constraint.	The system must never imply an action was executed without execution evidence.
13	Post-generation evidence and claim validation.	Fluent generation is not evidence of factual correctness.
14	Deflection rate excluded from the headline KPI.	Automation volume is not equivalent to customer resolution.
15	Production infrastructure deferred to Track B.	Distributed queues and databases add complexity without strengthening the take-home proof.

Table 10. Every non-obvious engineering decision made during the build, in the order they arose.

12. Conclusion

This project is intentionally not presented as a production-ready autonomous support agent. It is presented as an engineered, measurable foundation: a deliberately simple classifier surrounded by evidence-aware retrieval, grounded generation, deterministic trust controls, privacy handling, and explicit, itemised failure analysis.

The strongest outcome of this work is not any single score in Section 5. It is a system that makes its own uncertainty visible and provides a concrete, prioritised path — set out in Section 10 — from that uncertainty to either verified resolution or safe human ownership.

Submission references: Hiver SDE Intern Take-Home Assignment · AI-Powered Customer Support Agent — System Design & SRS v3.0 · AI Support Agent — Design Decisions / Architecture v1 · Kaggle Customer Support on Twitter (thoughtvector/customer-support-on-twitter).